LETTER OF TRANSMITTAL

HIGHLAND ENGINEERING & SURVEYING, INC.
1426 MEMORIAL DRIVE
OAKLAND, MD 21550
TELEPHONE: 301-334-6185
FAX: 301-334-8317

DATE 5/18/18

TO: Leslie Medice
lam@medicegroup.com

REFERENCE: 158 Laurel Lane
Proposed Hot Tub

FROM: Larry Bayles

MESSAGE: Additional support is needed under first floor deck columns and under the planned hot tub position. The garage beam needs significant reinforcement due to added weight estimated at 5086 lbs.

DOCUMENTS ENCLOSED:
(2) pg findings & recommendations
(2) pg. drawings
(7) pg. calculation sheets

HIGHLAND ENGINEERING & SURVEYING, INC.
1426 MEMORIAL DRIVE
OAKLAND, MARYLAND 21550
301-334-6185
(FAX) 301-334-8317

Medice – 158 Laurel Lane
Proposed Installation of a 6-Person Hot Tub
Larry Bayles May 18, 2018

Proposed Location

1. The hot tub is planned to be installed on a first floor deck that overlies a garage roof.
2. An attached approximate plan view, drawn using field measurements, shows the preferred position of the tub. Dimensions shown are 91"x91" as obtained from general product information for a 6-person tub.
3. The deck floor has seated columns that transfer loads to framing from roofing. The roof extends over about 8' of deck. Ends of some columns are not seated over joists and there has been compression of flooring in those areas due too weak support. As shown on the plan view, considered tub dimensions fit between two column locations. Observed joist positions generally are 2' apart and depth to the base was estimated to be 9".

Underlying Framing

1. Inside the garage there is a central beam that spans about 35' from wall to wall. Two 6"x6" wood columns support the beam which has ends seated on block. Ends were notched about 3".
2. The beam consists of (3) 2"x10". Plies are nailed together. Some butt joints occur between supports which creates a weakened condition, but the existing member appears to be in good condition. A metal ceiling covers framing above.
3. The concrete floor slab generally is solid around column bases which is indicative of good bearing capacity.

Evaluation of Framing

1. Roof loads for snow and dead weight at column locations on the first floor deck were calculated and applied to considered joist positions in addition to design loads of 40 PSF live and 10 PSF dead for the floor. Analysis determined that 3-ply 2x10 pressure treated southern pine is needed for those locations or 4-ply 2"x8"..
2. Using general information for a 6-person hot tub weight and holding capacity, it was determined that 2"x10" or 2-ply 2"x8" pressure treated southern pine floor joists at 12" center would be needed to support the considered design loads.
3. The interior garage beam was evaluated based on considered positions of loads from joists supporting the roof structure, members supporting the proposed hot tub

and other first floor deck framing carried by the existing beam. Calculation results indicate that reinforcement using two additional plies of laminated veneer lumber (LVL) is needed.

Recommendations

1. Install two plies of LVL to the existing beam. Though the left side is not affected by new hot tub loads, there is a large concentrated load from one first floor deck column and it is recommended that the entire beam be reinforced.
   a. See Drawing D-2 for details.
   b. Note that beam ends should be seated on flashing to prevent direct contact with wood on concrete blocks.
   c. Observe column bases during the work to identify any stress changes in concrete. If stresses are noted, contact the engineer.
2. Remove deck flooring and install joist reinforcement under existing columns so as to create members consisting of 3-ply 2"x10" No. 1 southern pine or 4-ply 2"x8" No. 1 southern pine. Provide a transfer path to carry loads to the garage beam below.
   a. Connect plies using two rows of ¼" x 4½" or ¼" x 6" heavy duty connectors at 18" centers in staggered top and bottom positions.
3. Remove deck flooring and install 2"x10" No. 1 southern pine at positions under the hot tub position so as to provide 12" spacing from center to center or install (2) 2"x8" to achieve the same support configuration. Connect plies with one row of ¼" x 3" heavy duty connectors. Provide a transfer path to carry loads to the garage beam below. All joist additions are to be pressure treated lumber.

STATE OF MARYLAND
LARRY GRAHAM BAYLES
No. 20877
REGISTERED
PROFESSIONAL ENGINEER

Larry Bayles, P.E.
5/18/18

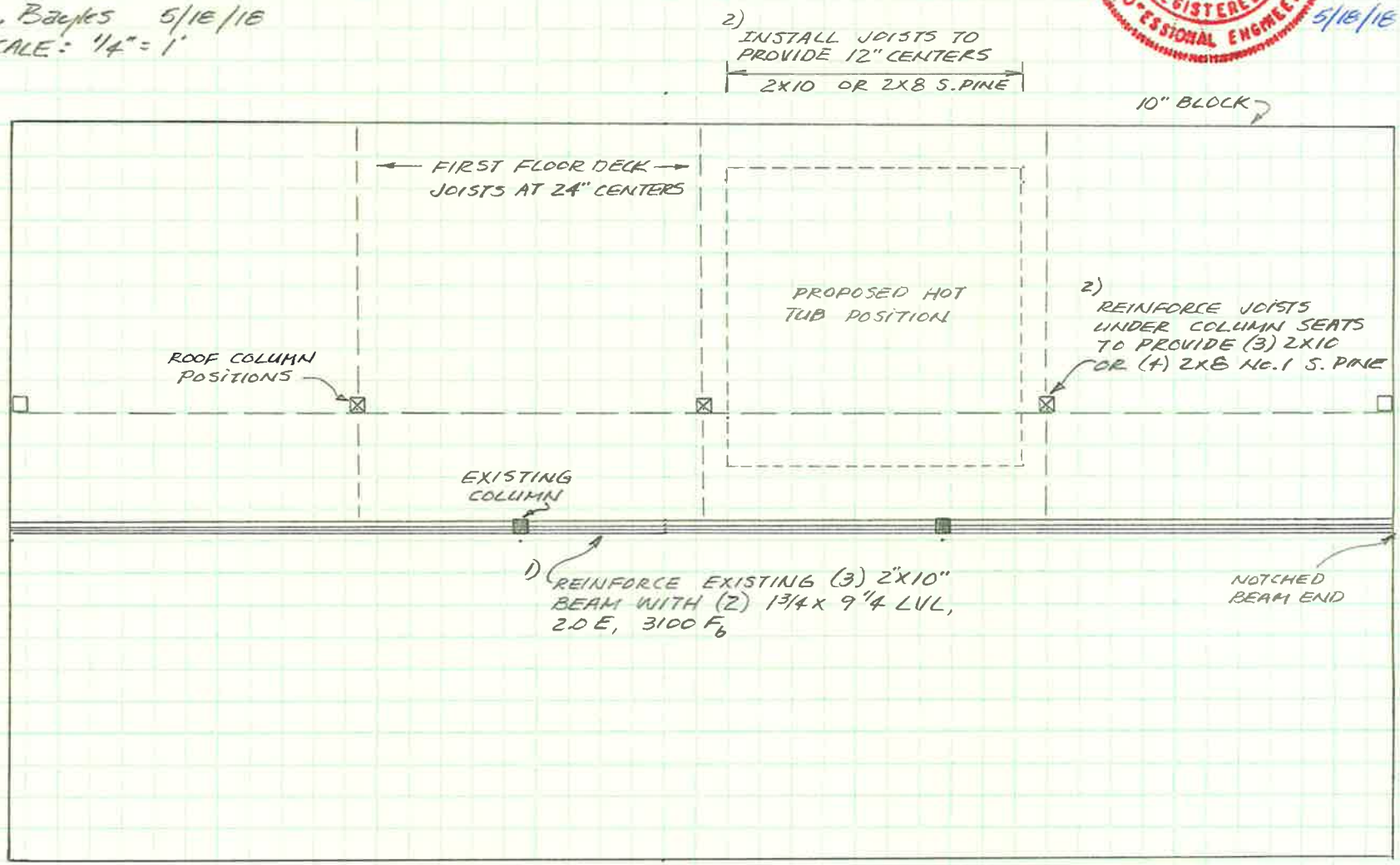
PROFESSIONAL CERTIFICATION

I hereby certify that these document components were prepared or approved by me and that I am a duly licensed Professional Engineer under the laws of the State of Maryland, License Number 20877 which expires June 24, 2019.

LGB/led

MEDICE – 158 LAUREL LANE
PROPOSED HOT TUB
"APPROXIMATE PLAN VIEW"
L. Bayles 5/18/18
SCALE: 1/4" = 1'

STATE OF MARYLAND
LARRY GRAHAM BAYLES
No. 20877
REGISTERED
PROFESSIONAL ENGINEER
5/18/18

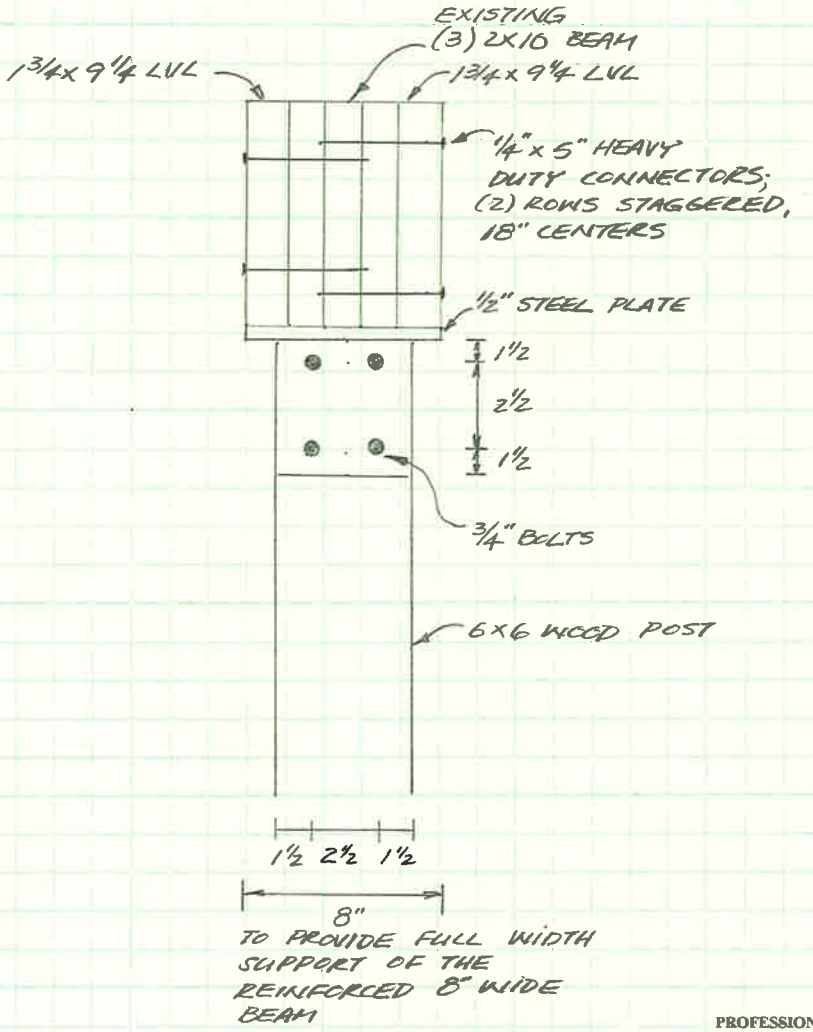


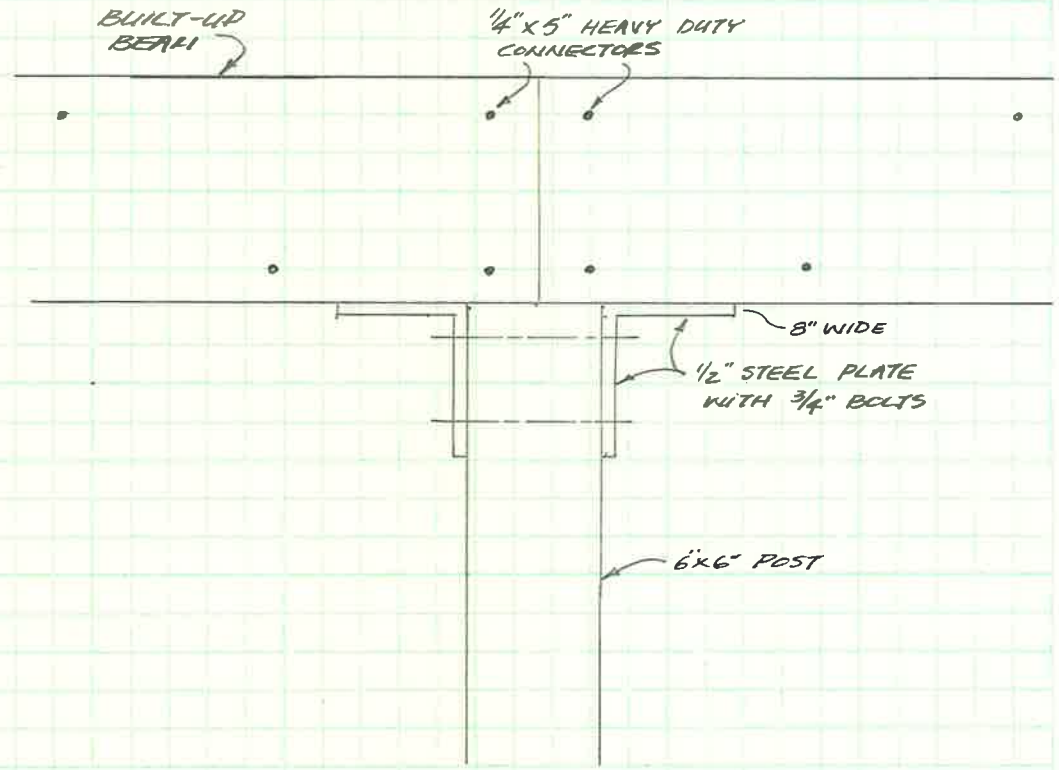
PROFESSIONAL CERTIFICATION
I hereby certify that these documents components were prepared or approved by me and that I am a duly licensed Professional Engineer under the laws of the State of Maryland, Licensed Number 20877 which expires June 24, 2019

NOTES:

1. SEE DRAWING D-2 FOR DETAILS ON REINFORCEMENT
2. PROVIDE TRANSFER PATHES TO CARRY LOADS FROM NEW FRAMING DOWN TO THE GARAGE REINFORCED BEAM; EXISTING CONFIGURATION UNDER DECK FLOOR IS UNKNOWN.

MEDICE - 158 LAUREL LANE
PROPOSED HOT TUB FRAMING REINFORCEMENT
L. Bayks 5/18/18
SCALE: 1½" = 1'





PROFESSIONAL CERTIFICATION
I hereby certify that these documents components were prepared or approved by me and that I am a duly licensed Professional Engineer under the laws of the State of Maryland, Licensed Number 208770 which expires June 24, 2019.

STATE OF MARYLAND
LARRY GRAHAM BAYLES
No. 20877
REGISTERED
PROFESSIONAL ENGINEER
5/18/18

#1

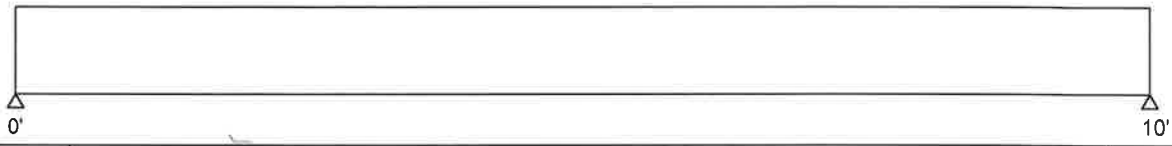
WoodWorks® SOFTWARE FOR WOOD DESIGN	COMPANY	PROJECT
		Medice - Hot Tub joists under roof load
	May 18, 2018 08:10:26	Beam2

Design Check Calculation Sheet

LOADS: (lbs, psf, or plf)

Load	Type	Distribution	Magnitude Start	Magnitude End	Location [ft] Start	Location [ft] End	Pattern Load?
1	Dead	Full UDL	20				No
2	Live	Full UDL	80				No
3	Snow	Point	2550		7.00		No
4	Dead	Point	892		7.00		No

MAXIMUM REACTIONS (lbs) and BEARING LENGTHS (in) :



Dead	368		724
Live	874		1785
Total	1241		2509
Bearing: Length	1.0		1.0

Lumber n-ply, S. Pine, No. 1, 2x10", 3-Plys

Lateral support: Top= full, Bottom= at supports; Repetitive factor: applied where permitted(refer to online help); Load combinations: ASCE 7-95

SECTION vs. DESIGN CODE NDS-1997: (stress=psi, and in)

Criterion	Analysis Value	Design Value	Analysis/Design
Shear	fv @d = 90	Fv' = 103	fv/Fv' = 0.87
Bending(+)	fb = 1391	Fb' = 1719	fb/Fb' = 0.81
Live Defl'n	0.14 = L/832	0.33 = L/360	0.43
Total Defl'n	0.23 = L/514	0.50 = L/240	0.47

o.k. for wet condition

ADDITIONAL DATA:

FACTORS:	F	CD	CM	Ct	CL	CF	CV	Cfu	Cr	LC#
Fb'+=	1250	1.15	1.00	1.00	1.000	1.04	1.000	1.00	1.15	4
Fv' =	90	1.15	1.00	1.00			(CH = 1.000)			4
Fcp'=	565		1.00	1.00						-
E' =	1.7 million		1.00	1.00						4

```
Bending(+): LC# 4 = D+S, M =  7438 lbs-ft
Shear     : LC# 4 = D+S, V = 2509, V@d = 2494 lbs
Deflection: LC# 4 = D+S   EI= 168.18e06 lb-in2/ply
Total Deflection = 1.50(Defln_dead) + Defln_Live.
  (D=dead  L=live  S=snow  W=wind  I=impact  C=construction)
  (All LC's are listed in the Analysis output)
```

DESIGN NOTES:

1. Please verify that the default deflection limits are appropriate for your application.
2. Sawn lumber bending members shall be laterally supported according to the provisions of NDS Clause 4.4.1.
3. BUILT-UP BEAMS: it is assumed that each ply is a single continuous member (that is, no butt joints are present) fastened together securely at intervals not exceeding 4 times the depth and that
each ply is equally top-loaded. Where beams are side-loaded, special fastening details may be required.

#2

WoodWorks® SOFTWARE FOR WOOD DESIGN	COMPANY	PROJECT
	May 18, 2018 08:10:49	Medice - hot tub joists under roof load Beam2

Design Check Calculation Sheet

LOADS: (lbs, psf, or plf)

Load	Type	Distribution	Magnitude Start	Magnitude End	Location [ft] Start	Location [ft] End	Pattern Load?
1	Dead	Full UDL	20				No
2	Live	Full UDL	80				No
3	Snow	Point	2550		7.00		No
4	Dead	Point	892		7.00		No

MAXIMUM REACTIONS (lbs) and BEARING LENGTHS (in) :

Dead	368		724
Live	874		1785
Total	1241		2509
Bearing: Length	1.0		1.0

Lumber n-ply, S. Pine, No. 1, 2x8", 4-Plys

Lateral support: Top= full, Bottom= at supports; Repetitive factor: applied where permitted(refer to online help); Load combinations: ASCE 7-95

SECTION vs. DESIGN CODE NDS-1997: (stress=psi, and in)

Criterion	Analysis Value	Design Value	Analysis/Design
Shear	fv @d = 86	Fv' = 103	fv/Fv' = 0.83
Bending(+)	fb = 1698	Fb' = 1984	fb/Fb' = 0.86
Live Defl'n	0.22 = L/534	0.33 = L/360	0.67
Total Defl'n	0.36 = L/330	0.50 = L/240	0.73

1.01 bending considered wet

ADDITIONAL DATA:

FACTORS:	F	CD	CM	Ct	CL	CF	CV	Cfu	Cr	LC#
Fb'+=	1250	1.15	1.00	1.00	1.000	1.20	1.000	1.00	1.15	4
Fv' =	90	1.15	1.00	1.00			(CH = 1.000)			4
Fcp'=	565		1.00	1.00						-
E' =	1.7 million		1.00	1.00						4

```
Bending(+): LC# 4 = D+S, M =  7438 lbs-ft
Shear     : LC# 4 = D+S, V = 2509, V@d = 2497 lbs
Deflection: LC# 4 = D+S   EI=  80.98e06 lb-in2/ply
Total Deflection = 1.50(Defln_dead) + Defln_Live.
  (D=dead  L=live  S=snow  W=wind  I=impact  C=construction)
  (All LC's are listed in the Analysis output)
```

DESIGN NOTES:

1. Please verify that the default deflection limits are appropriate for your application.
2. Sawn lumber bending members shall be laterally supported according to the provisions of NDS Clause 4.4.1.
3. BUILT-UP BEAMS: it is assumed that each ply is a single continuous member (that is, no butt joints are present) fastened together securely at intervals not exceeding 4 times the depth and that
each ply is equally top-loaded. Where beams are side-loaded, special fastening details may be required.

#3

WoodWorks® SOFTWARE FOR WOOD DESIGN	COMPANY	PROJECT
	May 18, 2018 06:59:10	Medice – tiot tub joists at 12" under tub (1) 2x10 Beam1

Design Check Calculation Sheet

LOADS: (lbs, psf, or plf)

Load	Type	Distribution	Magnitude Start	Magnitude End	Location [ft] Start	Location [ft] End	Pattern Load?
1	Dead	Full UDL	10				No
2	Dead	Partial UDL	80	79.80	1.00	8.58	No
3	Live	Partial UDL	40	40.00	0.00	1.00	No
4	Live	Partial UDL	100	100.00	1.00	8.58	No
5	Live	Partial UDL	40	40.00	8.58	10.00	No

MAXIMUM REACTIONS (lbs) and BEARING LENGTHS (in) :

	0'	10'
Dead	365	340
Live	437	418
Total	802	758
Bearing: Length	1.0	1.0

Lumber-soft, S. Pine, No. 1, 2x10"

Spaced at 16" c/c;

Lateral support: Top= full, Bottom= at supports; Repetitive factor: applied where permitted(refer to online help); Load combinations: ASCE 7-95

SECTION vs. DESIGN CODE NDS-1997: (stress=psi, and in)

Criterion	Analysis Value	Design Value	Analysis/Design
Shear	fv @d = 83	Fv' = 90	fv/Fv' = 0.92
Bending(+)	fb = 1272	Fb' = 1495	fb/Fb' = 0.85
Live Defl'n	0.13 = L/937	0.33 = L/360	0.38
Total Defl'n	0.30 = L/404	0.50 = L/240	0.59

ADDITIONAL DATA:

FACTORS:	F	CD	CM	Ct	CL	CF	CV	Cfu	Cr	LC#
Fb'+=	1250	1.00	1.00	1.00	1.000	1.04	1.000	1.00	1.15	2
Fv' =	90	1.00	1.00	1.00			(CH = 1.000)			2
Fcp'=	565		1.00	1.00						-
E' =	1.7 million		1.00	1.00						2

```
Bending(+): LC# 2 = D+L, M =  2267 lbs-ft
Shear     : LC# 2 = D+L, V =  802, V@d =  764 lbs
Deflection: LC# 2 = D+L   EI= 168.18e06 lb-in2
Total Deflection = 1.50(Defln_dead) + Defln_Live.
  (D=dead  L=live  S=snow  W=wind  I=impact  C=construction)
  (All LC's are listed in the Analysis output)
```

DESIGN NOTES:

1. Please verify that the default deflection limits are appropriate for your application.
2. Sawn lumber bending members shall be laterally supported according to the provisions of NDS Clause 4.4.1.

#4

WoodWorks® SOFTWARE FOR WOOD DESIGN	COMPANY	PROJECT
	May 18, 2018 07:00:28	Medice – hot tub joists @ 12" under tub (2) 2x8 Beam1

Design Check Calculation Sheet

LOADS: (lbs, psf, or plf)

Load	Type	Distribution	Magnitude Start	Magnitude End	Location [ft] Start	Location [ft] End	Pattern Load?
1	Dead	Full UDL	10				No
2	Dead	Partial UDL	80	79.80	1.00	8.58	No
3	Live	Partial UDL	40	40.00	0.00	1.00	No
4	Live	Partial UDL	100	100.00	1.00	8.58	No
5	Live	Partial UDL	40	40.00	8.58	10.00	No

MAXIMUM REACTIONS (lbs) and BEARING LENGTHS (in) :

Dead	365		340
Live	437		418
Total	802		758
Bearing: Length	1.0		1.0

Lumber-soft, S. Pine, No. 1, 4x8"

Spaced at 16" c/c;

Lateral support: Top= full, Bottom= at supports; Repetitive factor: applied where permitted(refer to online help); Load combinations: ASCE 7-95

SECTION vs. DESIGN CODE NDS-1997: (stress=psi, and in)

Criterion	Analysis Value	Design Value	Analysis/Design
Shear	fv @d = 46	Fv' = 90	fv/Fv' = 0.51
Bending(+)	fb = 887	Fb' = 1581	fb/Fb' = 0.56
Live Defl'n	0.11 = <L/999	0.33 = L/360	0.34
Total Defl'n	0.26 = L/454	0.50 = L/240	0.53

ADDITIONAL DATA:

FACTORS:	F	CD	CM	Ct	CL	CF	CV	Cfu	Cr	LC#
Fb'+=	1250	1.00	1.00	1.00	1.000	1.10	1.000	1.00	1.15	2
Fv' =	90	1.00	1.00	1.00			(CH = 1.000)			2
Fcp'=	565		1.00	1.00						-
E' =	1.7 million		1.00	1.00						2

```
Bending(+): LC# 2 = D+L, M =  2267 lbs-ft
Shear     : LC# 2 = D+L, V =  802, V@d =  772 lbs
Deflection: LC# 2 = D+L   EI= 188.95e06 lb-in2
Total Deflection = 1.50(Defln_dead) + Defln_Live.
  (D=dead  L=live  S=snow  W=wind  I=impact  C=construction)
  (All LC's are listed in the Analysis output)
```

DESIGN NOTES:

1. Please verify that the default deflection limits are appropriate for your application.
2. Sawn lumber bending members shall be laterally supported according to the provisions of NDS Clause 4.4.1.

WoodWorks® SOFTWARE FOR WOOD DESIGN

COMPANY	PROJECT
	medice hot tub framing garage beam - heaviest load
May 18, 2018 08:41:27	Beam1

Design Check Calculation Sheet

LOADS: (lbs, psf, or plf)

Load	Type	Distribution	Magnitude Start	Magnitude End	Location [ft] Start	Location [ft] End	Pattern Load?	
1	Dead	Point	50		0.00		No	1' joist space
2	Live	Point	200		0.00		No	" "
3	Live	Point	400		1.00		No	2' joist space
4	Dead	Point	100		1.00		No	
5	Dead	Point	100		3.00		No	same
6	Live	Point	400		3.00		No	
7	Live	Point	1785		4.00		No	joist point load from roof col.
8	Dead	Point	724		4.00		No	" " " " " "
9	Dead	Partial UDL	340	340.00	5.00	10.00	No	under hot tub at 12" spacing
10	Live	Partial UDL	418	418.00	5.00	10.00	No	" " " " "
11	Dead	Full UDL	40				No	front deck
12	Dead	Full UDL	160				No	front deck

MAXIMUM REACTIONS (lbs) and BEARING LENGTHS (in) :

	0'	10'
Dead	2069	2605
Live	2434	2441
Total	4503	5046
Bearing: Length	1.7	1.9

LVL n-ply, 2.0E, 3100Fb, 1-3/4x9-1/4", 2-Plys

Load combinations: ASCE 7-95

WARNING: point loads applied at support locations only affect maximum reactions and bearing lengths. The point loads have been added to the reactions without regard for load patterns.

SECTION vs. DESIGN CODE NDS-1997: (lbs, lbs-ft, or in)

Criterion	Analysis Value	Design Value	Analysis/Design
Shear	V @d = 4308	Vr = 6152	V/Vr = 0.70
Bending(+)	M = 13412	Mr = 13358	M/Mr = 1.00
Live Defl'n	0.27 = L/449	0.33 = L/360	0.80
Total Defl'n	0.63 = L/190	0.67 = L/180	0.95

ADDITIONAL DATA:

FACTORS:	F	CD	CM	Ct	CL	CF	CV	Cfu	Cr	LC#
Fb'+=	3100	1.00	1.00	1.00	1.000	1.04	1.000	1.00	1.00	2
Fv' =	285	1.00	1.00	1.00			(CH = 1.000)			2
Fcp'=	750		1.00	1.00						-
E' =	2.0 million		1.00	1.00						2

```
Bending(+): LC# 2 = D+L, M = 13412 lbs-ft
Shear     : LC# 2 = D+L, V = 5046, V@d = 4308 lbs
Deflection: LC# 2 = D+L   EI= 461.68e06 lb-in2/ply
Total Deflection = 1.50(Defln_dead) + Defln_Live.
  (D=dead  L=live  S=snow  W=wind  I=impact  C=construction)
  (All LC's are listed in the Analysis output)
```

DESIGN NOTES:

1. Please verify that the default deflection limits are appropriate for your application.
2. SCL-BEAMS (Structural Composite Lumber): the attached SCL selection is for preliminary design only. For final member design contact your local SCL manufacturer.
3. BUILT-UP SCL-BEAMS: contact manufacturer for connection details when loads are not applied equally to all plys.

#6

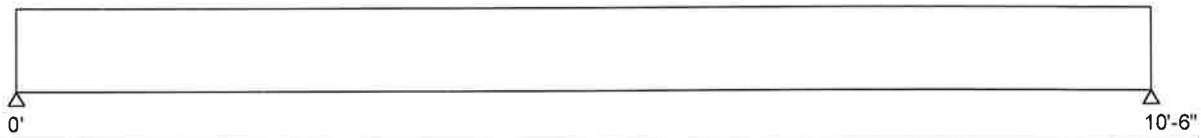
WoodWorks® SOFTWARE FOR WOOD DESIGN	COMPANY May 18, 2018 09:00:11	PROJECT medice hot tub framing garage beam - right side Beam1

Design Check Calculation Sheet

LOADS: (lbs, psf, or plf)

Load	Type	Distribution	Magnitude Start	Magnitude End	Location [ft] Start	Location [ft] End	Pattern Load?
1	Dead	Point	340		1.00		No
2	Live	Point	418		1.00		No
3	Live	Point	418		2.00		No
4	Dead	Point	340		2.00		No
5	Dead	Partial UDL	135	135.00	2.50	10.50	No
6	Live	Partial UDL	360	360.00	2.50	10.50	No

MAXIMUM REACTIONS (lbs) and BEARING LENGTHS (in) :



Dead	994		766
Live	1814		1902
Total	2808		2668
Bearing: Length	1.1		1.0

LVL n-ply, 2.0E, 3100Fb, 1-3/4x9-1/4", 2-Plys

Load combinations: ASCE 7-95

SECTION vs. DESIGN CODE NDS-1997: (lbs, lbs-ft, or in)

Criterion	Analysis Value	Design Value	Analysis/Design
Shear	V @d = 2808	Vr = 6152	V/Vr = 0.46
Bending(+)	M = 7190	Mr = 13358	M/Mr = 0.54
Live Defl'n	0.22 = L/581	0.35 = L/360	0.62
Total Defl'n	0.36 = L/351	0.70 = L/180	0.51

→ Shear o.k. for 3" notch condition checked with reduced depth

ADDITIONAL DATA:

FACTORS:	F	CD	CM	Ct	CL	CF	CV	Cfu	Cr	LC#
Fb'+=	3100	1.00	1.00	1.00	1.000	1.04	1.000	1.00	1.00	2
Fv' =	285	1.00	1.00	1.00			(CH = 1.000)			2
Fcp'=	750		1.00	1.00						-
E' =	2.0 million		1.00	1.00						2

```
Bending(+): LC# 2 = D+L, M =  7190 lbs-ft
Shear     : LC# 2 = D+L, V = 2808, V@d = 2808 lbs
Deflection: LC# 2 = D+L   EI= 461.68e06 lb-in2/ply
Total Deflection = 1.50(Defln_dead) + Defln_Live.
  (D=dead  L=live  S=snow  W=wind  I=impact  C=construction)
  (All LC's are listed in the Analysis output)
```

DESIGN NOTES:

1. Please verify that the default deflection limits are appropriate for your application.
2. SCL-BEAMS (Structural Composite Lumber): the attached SCL selection is for preliminary design only. For final member design contact your local SCL manufacturer.
3. BUILT-UP SCL-BEAMS: contact manufacturer for connection details when loads are not applied equally to all plys.

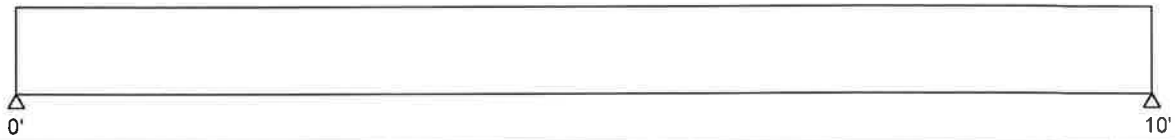
WoodWorks® SOFTWARE FOR WOOD DESIGN	COMPANY May 17, 2018 14:31:07	PROJECT medice hot tub framing joists without other loads Beam1

Design Check Calculation Sheet

LOADS: (lbs, psf, or plf)

Load	Type	Distribution	Magnitude Start	Magnitude End	Location [ft] Start	Location [ft] End	Pattern Load?
1	Dead	Full UDL	20				No
2	Live	Full UDL	80				No

MAXIMUM REACTIONS (lbs) and BEARING LENGTHS (in) :



Dead	100		100
Live	400		400
Total	500		500
Bearing: Length	1.0		1.0

Lumber-soft, S. Pine, No. 1, 2x10"

Spaced at 16" c/c;

Service: wet; Lateral support: Top= full, Bottom= at supports; Repetitive factor: applied where permitted(refer to online help); Load combinations: ASCE 7-95

SECTION vs. DESIGN CODE NDS-1997: (stress=psi, and in)

Criterion	Analysis Value	Design Value	Analysis/Design
Shear	fv @d = 46	Fv' = 87	fv/Fv' = 0.52
Bending(+)	fb = 701	Fb' = 1271	fb/Fb' = 0.55
Live Defl'n	0.12 = <L/999	0.33 = L/360	0.36
Total Defl'n	0.16 = L/733	0.50 = L/240	0.33

ADDITIONAL DATA:

FACTORS:	F	CD	CM	Ct	CL	CF	CV	Cfu	Cr	LC#
Fb'+=	1250	1.00	0.85	1.00	1.000	1.04	1.000	1.00	1.15	2
Fv' =	90	1.00	0.97	1.00			(CH = 1.000)			2
Fcp'=	565		0.67	1.00						-
E' =	1.7 million		0.90	1.00						2

```
Bending(+): LC# 2 = D+L, M =  1250 lbs-ft
Shear     : LC# 2 = D+L, V =  500, V@d =  423 lbs
Deflection: LC# 2 = D+L   EI= 168.18e06 lb-in2
Total Deflection = 1.50(Defln_dead) + Defln_Live.
  (D=dead  L=live  S=snow  W=wind  I=impact  C=construction)
  (All LC's are listed in the Analysis output)
```

DESIGN NOTES:

1. Please verify that the default deflection limits are appropriate for your application.
2. Sawn lumber bending members shall be laterally supported according to the provisions of NDS Clause 4.4.1.